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        NASA-GLENN CHEMICAL EQUILIBRIUM PROGRAM CEA2, FEBRUARY 5, 2004
                  BY  BONNIE MCBRIDE AND SANFORD GORDON
      REFS: NASA RP-1311, PART I, 1994 AND NASA RP-1311, PART II, 1996

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 ### CEA analysis performed on Fri 07-Mar-2025 08:24:06

 # Problem Type: "Rocket" (Infinite Area Combustor)

 prob case=_______________4066 ro equilibrium

 # Pressure (1 value):
 p,bar= 11
 # Supersonic Area Ratio (3 values):
 supar= 121, 135, 80

 # You selected the following reactants:
 reac
 name N2H4(L)           wt%=100.0000  t,k= 579.350

 # You selected these options for output:
 # short version of output
 output short
 # Proportions of any products will be expressed as Mass Fractions.
 output massf
 # Heat will be expressed as siunits
 output siunits
 # Transport properties calculated
 output transport


 # Input prepared by this script:/var/www/sites/cearun.grc.nasa.gov/cgi-bin/CEARU
 N/prepareInputFile.cgi

 ### IMPORTANT:  The following line is the end of your CEA input file!
 end

              THEORETICAL ROCKET PERFORMANCE ASSUMING EQUILIBRIUM

           COMPOSITION DURING EXPANSION FROM INFINITE AREA COMBUSTOR

 Pin =   159.5 PSIA
 CASE = _______________

             REACTANT                    WT FRACTION      ENERGY      TEMP
                                          (SEE NOTE)     KJ/KG-MOL      K
 NAME        N2H4(L)                      1.0000000     81585.966    579.350

 O/F=    0.00000  %FUEL=  0.000000  R,EQ.RATIO= 0.000000  PHI,EQ.RATIO= 0.000000

                 CHAMBER   THROAT     EXIT     EXIT     EXIT
 Pinf/P            1.0000   1.8715  2176.37  2475.05  1335.18
 P, BAR            11.000   5.8778  0.00505  0.00444  0.00824
 T, K             1200.00  1017.58   317.79   314.64   330.42
 RHO, KG/CU M    1.1784 0 7.4279-1 2.4917-3 2.2245-3 3.8289-3
 H, KJ/KG         2545.97  2006.55  -728.06  -753.95  -625.99
 U, KJ/KG         1612.52  1215.24  -930.91  -953.74  -841.16
 G, KJ/KG        -17443.1 -14943.9 -6021.66 -5995.07 -6130.04
 S, KJ/(KG)(K)    16.6576  16.6576  16.6576  16.6576  16.6576

 M, (1/n)          10.689   10.692   13.026   13.094   12.768
 (dLV/dLP)t      -1.00066 -1.00096 -1.11050 -1.11141 -1.10620
 (dLV/dLT)p        1.0037   1.0062   2.9393   2.9720   2.8032
 Cp, KJ/(KG)(K)    3.0021   2.9435  24.0731  24.4999  22.3278
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GAMMAs            1.3520   1.3634   1.1345   1.1331   1.1402
SON VEL,M/SEC     1123.4   1038.7    479.7    475.8    495.3
MACH NUMBER        0.000    1.000    5.334    5.399    5.085

TRANSPORT PROPERTIES (GASES ONLY)
  CONDUCTIVITY IN UNITS OF MILLIWATTS/(CM)(K)

VISC,MILLIPOISE  0.41492  0.37206  0.15075  0.14908  0.15735

 WITH EQUILIBRIUM REACTIONS

Cp, KJ/(KG)(K)    3.0021   2.9435  24.0731  24.4999  22.3278
CONDUCTIVITY      2.7026   2.3779   5.5829   5.6594   5.2692
PRANDTL NUMBER    0.4609   0.4606   0.6500   0.6454   0.6668

 WITH FROZEN REACTIONS

Cp, KJ/(KG)(K)    2.9859   2.9118   2.3483   2.3374   2.3903
CONDUCTIVITY      2.6940   2.3631   0.7294   0.7158   0.7835
PRANDTL NUMBER    0.4599   0.4585   0.4853   0.4868   0.4800

PERFORMANCE PARAMETERS

Ae/At                      1.0000   121.00   135.00   80.000
CSTAR, M/SEC               1425.8   1425.8   1425.8   1425.8
CF                         0.7285   1.7948   1.8018   1.7666
Ivac, M/SEC                1800.5   2638.2   2646.8   2604.1
Isp, M/SEC                 1038.7   2558.9   2569.0   2518.7

MASS FRACTIONS

*H2              0.12563  0.12554  0.07487  0.07366  0.07956
NH3              0.00106  0.00153  0.28695  0.29375  0.26054
*N2              0.87332  0.87293  0.63818  0.63259  0.65991

 * THERMODYNAMIC PROPERTIES FITTED TO 20000.K

NOTE. WEIGHT FRACTION OF FUEL IN TOTAL FUELS AND OF OXIDANT IN TOTAL OXIDANTS
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